For Examiner's Use

5 Fig. 5.1 shows a demonstration of a *stationary wave* on a string.

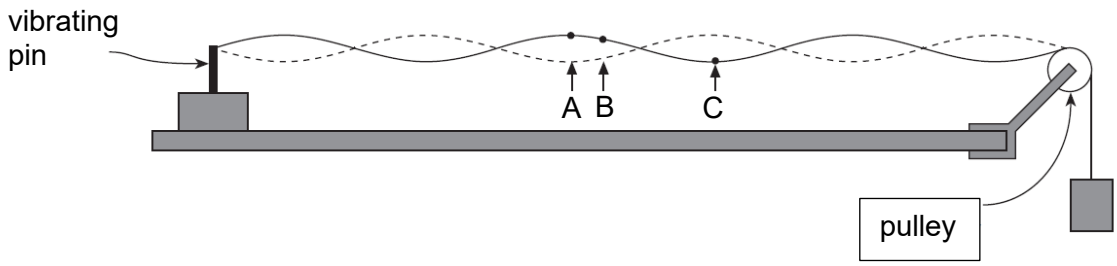

Fig. 5.1

(a) **(i)** State whether the following pairs of points on the string are vibrating *in phase*, *in antiphase*, or neither in phase nor in antiphase.

A and B ……

A and C …….. [2]

(ii) Describe briefly how you could check this experimentally.

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…… [1]

(b) **(i)** Progressive waves transfer energy through the medium; stationary waves do not do this. Describe the difference in term of the amplitude of the wave between progressive and stationary waves.

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…… [2]

(ii) Explain how, in the set-up as shown in Fig. 5.1, the stationary wave can be thought of as arising from progressive waves.

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..... [2]